WACV 2026 Applications Track — Peer Review Report

Paper #0000: Viewpoint Invariance Is a Theorem, Not a Fit: An SE(3)-Equivariant Recurrence for Rehabilitation Exercise Assessment

Recommendation: Weak Accept *(borderline; leaning accept)*	**Confidence Rating:** 4 / 5 *Adjacent to equivariant models/skeleton assessment; non-clinician.*
Ethics Flags: None	**Reproducibility:** Above Average (released certificates, hash-locked inputs)

PART 1: FORMAL REVIEW FORM

1. Paper Summary

The paper reframes skeleton-based rehabilitation exercise assessment around robustness rather than clean-data accuracy. Its central empirical claim is that on clean frontal KIMORE the aggregate metric is saturated: an SE(3)-equivariant recurrent model (EGRU), a point-cloud transformer (PCT), and a GRU on hand-crafted invariants tie within a measured 0.33 MAD nondeterminism floor. The paper argues the decisive axis is therefore off-frontal, and contributes: (i) a protocol audit (an 18.6% epoch-selection inflation, the nondeterminism floor, a gated pooled protocol); (ii) exact viewpoint invariance by construction (Prop. 1), certified by eight numerical gates and validated on real NTU RGB+D Cross-View; (iii) a measured justification for the steerable encoder via graceful sensor-node-failure degradation, including a lighter-E(n) comparison behind the identical invariant cut; (iv) a second-corpus replication (REHAB24-6); (v) a deployment study (int8 precision budget, causal streaming at 112 fps) and two reported negative results.

2. Strengths

- **S1 — Methodological Hygiene:** Evaluation methodology is well above the norm for this literature. Establishing a nondeterminism floor before asserting any accuracy difference, and gating every robustness curve on a per-exercise mean-predictor floor, are correct and rarely done. The paper is right that a degradation curve without a floor is uninterpretable, and the field would benefit from adopting both practices.
- **S2 — Transferable Insight:** The aggregate-vs-per-sequence distinction is a genuine and transferable insight. Showing that rotation augmentation flattens the mean while individual patient scores still swing 3.03 MAD (of 50) is the sharpest result in the paper. It is a real methodological warning with implications beyond this application.
- **S3 — Commendable Honesty:** Unusual and commendable honesty. The paper openly reports that it loses clean accuracy to the hand-crafted baseline, that PCT is the most node-robust model, that canonicalization ties it offline, and it gives two negative results equal prominence. Certificates are falsifiable and re-runnable. This is the right posture for an applications venue.

- **S4 — Deployment Realism:** Deployment realism. The int8 activation-vs-weight decomposition, the honest time-to-first-reliable-score, and reporting GPU latency as an upper bound (rather than overclaiming edge performance) are careful and credible.

3. Weaknesses

- **W1 — Missing Clinical Anchor:** The clinical significance of the headline quantity is never established (critical for this track). The paper's motivating promise is 'the same patient the same score from a different room,' yet it never states what change on the 50-point scale alters a clinical decision, nor the inter-rater reliability (clinician-vs-clinician MAD) on these instruments. Without this anchor, the reader cannot judge whether 3.03 MAD is a clinically dangerous swing or a rounding curiosity, nor whether 9×10^{-6} is meaningfully better than the InvariantGRU's exact 0 in any way a patient would experience.
- **W2 — Narrow Single-Cell Niche:** The authors' own honesty exposes a single-cell niche that the paper underplays. On every individual axis, something beats EGRU: clean accuracy (InvariantGRU 6.31 < 6.73), viewpoint (InvariantGRU exactly 0; Canon-PCA ≈ 0), node failure (PCT +2.27 < +3.76). The actual claim reduces to one intersection: EGRU is the only model that is both certified worst-case viewpoint-invariant and clears the node-failure floor at two dead joints (Table 6 supports this). That is real, but its entire weight rests on 'certified worst-case,' and the paper does not sufficiently confront why a deployer facing only azimuth sweeps and occasional dropout wouldn't choose the 0.21M InvariantGRU or Canon-PCA+PCT.
- **W3 — Self-Built Comparison Set:** The saturation argument is made largely over the authors' own constructions. The comparison set (PCT, PCT+aug, InvariantGRU, EGNN, Canon-PCA) is self-built. Published KIMORE results ([3–5]) are cited but their numbers never enter a table. The claim 'accuracy is saturated so external comparison is moot' is somewhat circular: reviewers cannot grant saturation until the published methods are shown to lie inside the band.
- **W4 — Statistical Rigor Discrepancy:** The pooled main result is not held to the same statistical rigor as the audit slice. The single-exercise audit rigorously reports Δ-vs-floor with a paired subject-cluster bootstrap and CI ($\Delta = -0.44$, clears zero in 1/3 seeds). Table 3 clears the floor numerically for the pooled task, but I do not see the equivalent paired-bootstrap CI for the pooled main result. Reporting the rigorous interval only where it looks unflattering, and a bare mean where it looks fine, invites suspicion.
- **W5 — Framing vs. Theoretical Novelty:** Novelty is in the evaluation, not the method — and the framing risks miscuing the grade. Prop. 1 is a standard result (scalar contractions of irreps are Wigner-D-invariant), and the invariant-projection read-out is established e3nn-family machinery, honestly cited. The E1–E8 certificate mostly confirms that a correctly-built invariant is invariant to roundoff. The 'theorem' framing risks inviting a methods-contribution grading the paper cannot win.
- **W6 — Thin Replication Corpus:** Second corpus is thin. REHAB24-6 has 10 subjects. It is adequate as replication of the structural viewpoint property, but too small to lean on heavily; the paper should not oversell it.
- **W7 — Presentation & Compilation:** Presentation and typesetting. The prose is extremely dense and rhetorically combative. More concretely, in the reviewed version Eq. (1) appears mangled and figure/line-number artifacts collide. Please verify the camera-ready-quality build.

4. Questions for the Authors (Rebuttal Period)

- What is the inter-rater MAD (or a published clinical decision threshold) on the KIMORE / REHAB24-6 scales? At what score change does a clinical recommendation flip? (Resolves W1)

- Please provide the pooled main-result Δ-vs-mean-predictor-floor with the paired subject-cluster bootstrap CI, matching the rigor of Table 2. (Resolves W4)
- Where do the published KIMORE methods [3–5] fall relative to the 0.33 MAD floor? Citing their reported numbers in Table 3 would substantiate saturation. (Resolves W3)
- Under what concrete deployment conditions is a worst-case certificate required rather than an empirically-flat canonicalization frame? (A named scenario would strengthen W2)
- Is the streaming −3.2 pt NTU accuracy delta within or outside your nondeterminism floor for that task?

5. Justification for Rating

This is a rigorous, honest, well-instrumented applications paper whose real contribution is an evaluation reframing — the nondeterminism floor, the mean-predictor gating, and the aggregate-vs-per-sequence degradation distinction — plus a narrow but genuine certified niche. Those are worth putting before the community, and the candor is a model for the subfield.

It is not a safe accept. The paper's greatest vulnerability is self-inflicted: by honestly showing that cheaper models tie or beat it on every individual axis, and by never anchoring its headline advantage in clinical units, it hands a reviewer a defensible reject narrative ('the authors themselves show their method usually isn't necessary, and never quantify when it is'). Whether this paper is accepted will likely turn on the rebuttal — specifically on W1 (clinical anchor) and W4 (pooled CI). If the authors supply the clinical-significance anchor and the pooled CI in rebuttal, I would raise to Accept. Absent those, I hold at Weak Accept and would not object to an AC decision either way.

PART 2: EDITORIAL ANALYSIS & REBUTTAL STRATEGY

> ### Key Editorial Assessment & Primary Action Items
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> The paper's greatest strength and its greatest vulnerability are identical: its extreme transparency. By demonstrating that simpler baselines win on individual metrics, the authors risk handing hostile reviewers a grounds for rejection. To elevate this paper to a definitive Accept, the authors must execute two critical rebuttal steps:
> 1. Anchor the 3.03 MAD swing to actual clinical decision thresholds / inter-rater variability.
> 2. Provide explicit bootstrap confidence intervals for the pooled main results in Table 3.

What's Genuinely Strong

- **Methodological Hygiene:** Measuring a nondeterminism floor before making accuracy claims, gating curves on a mean-predictor floor, and separating per-sequence degradation from aggregate accuracy are exemplary practices.
- **Transferable Insight:** The aggregate-flatness vs. per-sequence-swing distinction (3.03 MAD hiding inside a flat mean) is the sharpest idea in the paper and acts as a vital cautionary result.
- **Honest Reporting:** Acknowledging loss of clean accuracy to hand-crafted GRU, PCT's superior node-robustness, and offline canonicalization ties demonstrates scientific integrity appropriate for an applications track.

Critical Pushback & Risk Areas

- **1. Narrow Single-Cell Advantage:** EGRU is beaten on clean accuracy (InvariantGRU), viewpoint (InvariantGRU/Canon-PCA), and node failure (PCT). The win exists solely at the single intersection of certified worst-case viewpoint invariance AND node-failure tolerance at 2 dead joints. The paper must explicitly justify why this niche matters.
- **2. Lack of Clinical Grounding:** The central premise ('same patient, same score from a different room') lacks clinical translation. Without establishing inter-rater MAD or clinical decision thresholds, 3.03 MAD cannot be evaluated as either dangerous or negligible.
- **3. Low Architectural Novelty:** Proposition 1 relies on standard scalar contractions of irreps (Wigner-D invariance). Framing must clearly emphasize evaluation protocol and certified niche rather than claiming novel theoretical machinery.
- **4. Self-Built Comparison Set:** Excluding published KIMORE SOTA numbers from summary tables weakens the saturation claim. Published SOTA metrics must be added to Table 3.
- **5. Pooled Result Integrity:** The single-slice audit reports paired subject-cluster bootstrap CIs, but the pooled main result omits them. Alignment across both analyses is necessary to prevent suspicion of selective reporting.
- **6. Thin Replication Corpus:** REHAB24-6 contains only 10 subjects. While suitable as a secondary check, claims derived from it should remain strictly framed as structural property replication.
- **7. Formatting & Rendering:** Dense prose, aggressive rhetoric, and mangled mathematical formulas (e.g., Eq. 1) cost technical credibility and require immediate cleanup for camera-ready submission.

Summary of Model Trade-offs Across Evaluated Axes

Model	Clean Accuracy (MAD)	Viewpoint Error	Node Failure Robustness	Certified Worst-Case?
EGRU (Proposed)	6.73	9×10^{-6}	Clears floor (+3.76)	YES (Prop. 1)
InvariantGRU	6.31 (Best)	0.0 (Exact Best)	Fails floor	No
PCT	6.55	Vulnerable without aug	+2.27 (Best)	No
Canon-PCA + PCT	6.58	≈ 0.0 (Empirical)	Degrades off-axis	No